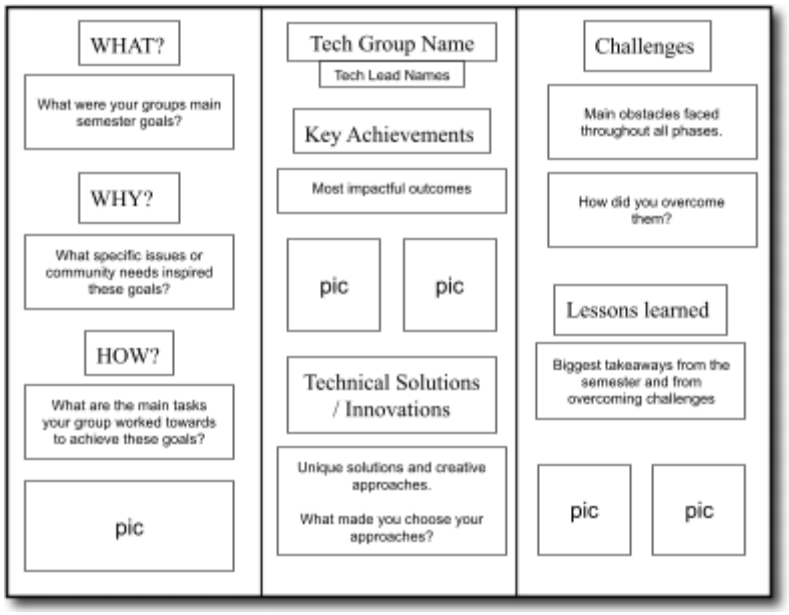


What

- The goal of Water Health & System Solutions was to develop a water filtration system that is compatible with the water catchment system in order to distribute clean and safe water to the community.

Why

- The community's current water source is a stream that is severely contaminated by sediment and other pollutants. Chlorine or other chemical based solutions could be used to filter bacteria out of the water, but they would not be able to remove the sediment. Thus, a new approach to collecting and purifying the water had to be developed.

How

- We developed a filtration system by creating a double bucket system for the rainwater to go through. The top bucket has layered filtration material (activated charcoal, gravel, and sand) that the water goes through to filter out sediment and other contaminants. Then the clean water flows into the second bucket and is stored until community members choose to get water.

Key Achievements:

- We developed and designed a dual-system water filter consisting of a first flush and geo-filter which was attached to a rain catchment device. Our design will be introduced to the community this summer on an assessment trip, where we will get more information about the feasibility of our design and how we can improve it to work better with the community.

Technical solutions/innovations

- Our design consists of a first flush system which is a pre-rinse cycle that prevents dirt, dust, debris, or other contaminants that are on the roof from entering the main water filter. The water flows into a geo-filter consisting of gravel, sand, and activated charcoal before collecting in a tank where the community can access the water using a tap.

Challenges

- After prototyping the geo-filter we discovered that the activated charcoal and gravel needed to be rinsed prior to being added to the filter. Rinsing the materials prevents dust or other small amounts of sediment from contaminating the water as it is filtered.
- The timeline for designing the first flush filter was rushed, making its design under-developed, so we were unable to complete the design we had planned.

Future Plans

- In our prototype for the first flush, the water is contained inside the pipe by duct tape, but in a more developed design there would be a stop plug that can be easily removed to release the water that accumulates in the first flush.
- We noticed a small amount of remaining sediment after tests with our geofilter, so we plan on doing more research about how to remove these suspended solids and the potential for a spring box in our design.